

Personalized Email Agent in School Context Using Lightweight LLM

1. Introduction

Email overload is a serious problem in modern life. Users receive a large number of emails including critical work notifications, personal communications, as well as those news subscriptions, marketing advertisements. Categorizing these emails manually is time-consuming and labor-intensive. What’s more, it is easy to miss important information. In school life, email is the most crucial communication channel connecting the school and students. Students and teachers receive a large amount of information, including: administrative announcements, course logistics, research and cooperation exchanges, service notices, transactional emails, communications, marketing promotions and spam. Therefore, users need an intelligent email assistant to help them with classification and keyword extraction.

Manually sorting these emails will increase cognitive and time burdens, and important information is often omitted or delayed. Therefore, the development of large model languages in email classification is very necessary, which currently has implied many applications. For example, use a large language model (LLM) to do text classification (Zhang et al., 2025), and prompt-based method for name entity recognition (Ashok et al., 2023). Moreover, recently some commercial products (e.g. Outlook) have begun integrating LLMs into email systems to provide users with a summarization function. Although commercial platforms are increasingly offering general AI summaries, they typically lack fine-grained, user-controlled interaction and often rely on large proprietary models, posing privacy and cost issues. Our work will feature user personalization in email labels and handling actions while proposing a low-cost solution using a lightweight model that is easy to deploy.

We address this gap with a personalized email assistant for school context that runs on a lightweight, zero-cost 10–15B LLM in on-device or self-hosted deployments. It performs user-defined multi-label classification, structured information extraction, and executing user-defined rules. The system integrates campus-aware retrieval for retrieval-augmented grounding, delivers privacy-preserving, low-latency operation without recurring application programming interface (API) costs.

2. Related Work

Currently, tool-like LLM based assistants use API to provide service. The performance of the AI tools highly rely on the performance of upstream LLM models. If the developers of the AI tools want to improve their service quality, they might have to choose a better performance LLM model. This kind of update strategy results in higher maintenance cost. Lightweight LLM with a small number of trained parameters provides an option for AI tools developers since its low cost. Prompt engineering and task-specific formatting help align outputs with label taxonomies while avoiding costly fine-tuning. However, due to its number of trained

parameters (roughly 10B to 20B parameter) is much smaller than normal LLM (larger than 150B), its overall reasoning ability and following instruction ability is not so good.

To enhance performance of light weight LLM in certain tasks, one method is to use post-pretrain and fine tuning. Through these two stages, LLM can learn context from the training data, and improve the instruction following capability of certain specific tasks. Distillation is another method for improving lightweight models' performance, reducing model size while retaining performance, enabling on-device use. The lightweight model acts as a student model, learning the reasoning ability from a larger model which acts as a teacher model.

Based on our observation, the current methods are constrained by two principal challenges. Firstly, they exhibit an intensive dependence on large-scale, high quality data. Processes such as post pre-train demand billions of tokens of curated text alongside thousands of instructional prompts set, while supervised fine-tuning requires thousands of well-labelling data with enough variety. Knowledge distillation similarly depends on extensive datasets for effective training. In specific application contexts like a school email system, data at this scale presents significant budgetary hurdles and raises data privacy concerns. Secondly, a full-parameter fine-tuning approach applied to a lightweight Large Language Model often leads to severe overfitting. For instance, we fine tuned a model using supervised fine tuning (SFT) with LoRA approach, and the training process used a classifier dataset (*High-Accuracy-Email-Classifier*, 2025) combined with system prompts. The outputs of the fine-tuned model coverage strictly to the standard email categories and cannot get expected result if we introduce a new category label. This declines the model's inherent capacity for generalization, causing it to fail in generating responses or content beyond the scope of its training data. This limitation fundamentally contradicts our core objective of developing a flexible, personal system capable of handling diverse and unpredictable situations.

To overcome these critical limitations, a more efficient and sustainable approach of applying LLM should be considered. Therefore, our work employs knowledge enhancement related method, including Chain-of-Thought (CoT) and Retrieval-Augmented Generation (RAG), which guide the model's reasoning process and expand the confines of the model's knowledge. While reducing the reliance on high-quality training data, our work can still offer users personalized choices.

3. Methodology

3.1 Overall Architecture

We adopt a multi-agent system with both parallelism and a controlled serial chain as the structure of email assistant to decompose email triage into tractable subtasks while preserving end-to-end coherence. Its design thinking is to break a large task into small subtasks and

maintain primary processing logic, decreasing overall inference time while preserving coupling along the whole task. Rather than overloading a single prompt with all objectives, we split the workflow into ingestion, retrieval, categorization and intent inference, structured information extraction, consolidation, and rule execution. This arrangement ensures that each agent for the subtasks can fulfill its intended function, without compromising performance due to excessive difficulty that would hinder lightweight models from generating satisfactory outcomes.

Our work is using the ERNIE-speed model, a product in 2024, featuring in its zero cost and light weight deployments, which perfectly matches our purpose of building a low-cost, lightweight agent. This model achieves sub-second median latency on commodity GPUs and remains feasible on CPU-only self-hosted setups, which satisfy the goal of application in on-device or self-hosted deployments.

Fig. 1 shows the high-level, sequential pipeline for processing information. The pipeline follows a parallel and serial high-level design. An incoming email is first normalized and passed through language determination to choose templates and local rules. Then based on the users' customized choices, importance scoring, label classification, summarization, information extraction, and potential response writing these five subtasks run in parallel.

While RAG supplies campus context to the information extraction to correct course codes, dates, locations, and aliases. Each subtask outputs constrained structured fragments that converge in a consolidation stage, which resolves conflicts. At the same time, the response component can generate optional auto-replies using user policies. After all, package everything into a standardized output. This design handles steps like retrieving information and preprocessing right at the beginning. It runs certain tasks in parallel to reduce inference waiting time (e.g. Time To First Token). Then, it uses retrieved knowledge and reasonable prompt engineering to make sure the final output is both accurate and coherent from start to finish.

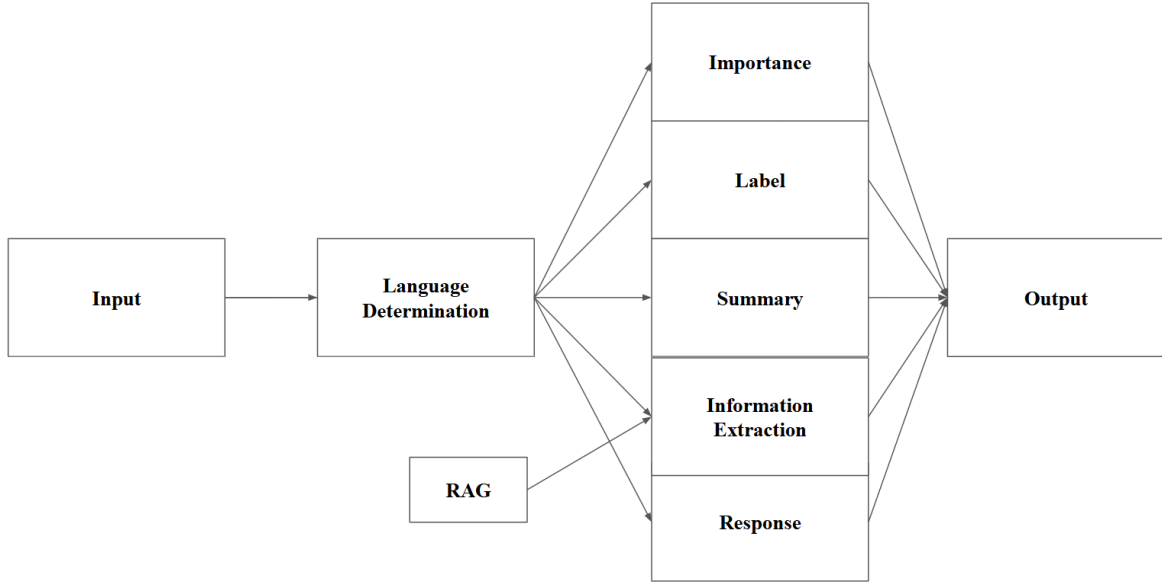


Fig. 1. High-level structure

3.2 Chain-of-Thought

Chain-of-Thought (CoT) prompting is an effective method to enhance reasoning capabilities in subtask by structuring the thinking process into clear steps. Starting from the description of different labels, CoT guides the inference of label category through systematic reasoning paths. For instance, given the body part of email, CoT enables the model to output more accurate label category results. This approach ensures logical output, alignment with the users' basic requirement and enhances the model's ability to handle users' customized demand. Incorporating CoT prompting strengthens reasoning precisely where lightweight models tend to falter, such as ambiguous multi-label assignments, intent disambiguation, and temporal normalization.

3.3 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) offers a powerful approach to enhance information extraction reasoning in email assistants by integrating external knowledge retrieval with generative language models. This method addresses the limitations of lightweight model low instruction-following capacity and low sensitivity to certain types of information in email by leveraging external data to augment the inference process, thereby improving the quality of output.

We construct a campus knowledge base, segment documents into 600 word chunks, and vectorize them with a embedding model. At query time, we perform hybrid retrieval: a lexical stage captures exact identifiers and codes, while a dense vector search recovers semantic

paraphrases and aliases. Our method uses hybrid RAG combined with a re-ranking model to rerank and select top 6 matched context chunks for LLM reference.

4. Experiments & Results

4.1 Effect of CoT

As the comparative result shown in Table 1, after using CoT, the labelling accuracy significantly improves and the outcomes are close to human expectation. Overall, using CoT leads to more accurate and context-aware labeling, especially for emails that might otherwise be misclassified as "advertisement." For example, "University Microsoft Storage Adjustment" and "Responsible AI Workshop" are correctly relabeled as "study" with CoT, recognizing their academic relevance. Similarly, "Lost card found in AIMS5703" shifts from "advertisement" to "study," reflecting its campus-related nature. However, in cases like "Offshore internship opportunities," CoT changes the label from "advertisement" to "career," suggesting it better captures the intent behind the message. The results show that CoT helps the system understand nuance and context, reducing over-reliance on surface keywords and improving label consistency with the actual purpose of the email.

Email Content	Label Without CoT	Label With CoT
Reminder of CTE(2)	study	study
University Microsoft Storage Adjustment	advertisement	study
Offshore internship opportunities	advertisement	career
HKSTP Tech FYP Sponsorship Programme	advertisement	advertisement
Responsible AI Workshop	advertisement	study
Reminder of CTE(1)	advertisement	study
Distinguished lecture announcement	career	career
Lost card found in AIMS5703	advertisement	study
AIMS5701 project requirement	study	study
AIMS5703 grade release	career	study

Table 1. Comparative results of label outputs with and without CoT

4.2 Effect of RAG

Fig. 2 illustrates the comparative effectiveness of two prompt engineering approaches—one without Retrieval-Augmented Generation (RAG) and the other incorporating RAG-in the context of extracting information in school email. The results demonstrate that including RAG significantly enhances the model's ability to extract detailed information. In the "Prompt without RAG" approach, the model struggles to analyze what is detailed information and output a brief summary about email body content. This demonstrates the limitations of relying solely on the model's internal knowledge, as it fails to understand fully about the task. Conversely, the "Prompt with RAG" approach leverages external knowledge retrieval to supplement the model's reasoning process. By integrating the knowledge base about key information in school email, the model can systematically extract relevant keywords. For example, the model correctly extracts course names, professor names, and specific time, demonstrating improved ability in extracting key information in school email context.

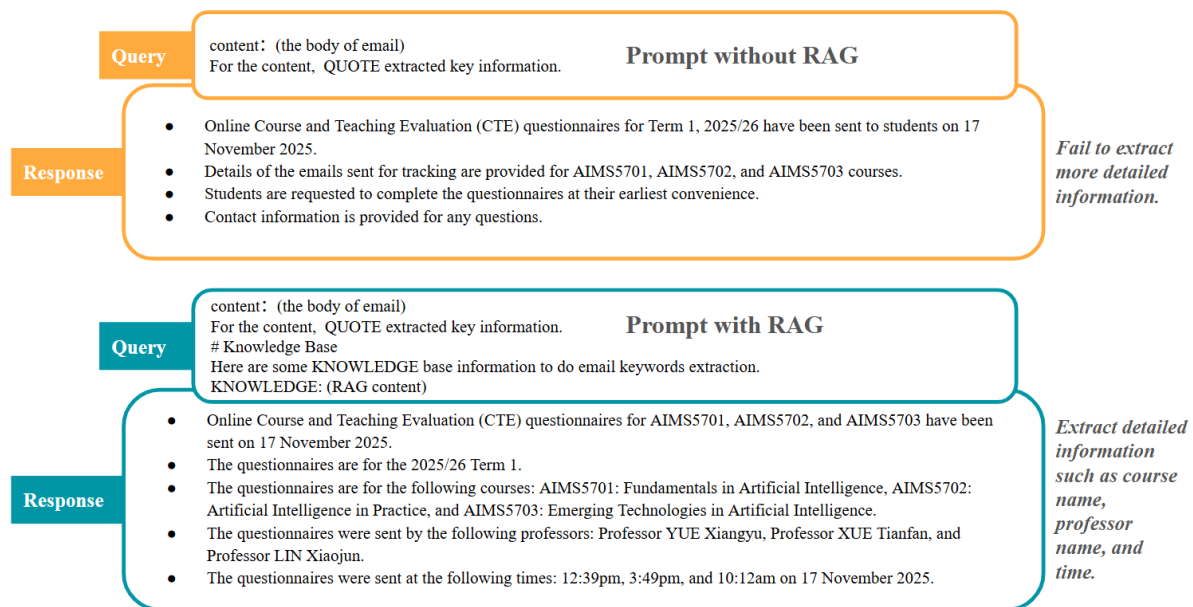


Fig. 2. Comparison of key information extraction with and without RAG

4.3 User Interface Demonstration

The graphical user interface of our email assistant is designed with a two-mode layout to support both interactive single-mail processing and automated batch operations, as shown in Fig. 3 and Fig. 4, respectively.

In the normal operation mode, users can directly input the content of an email into a provided text field. The interface allows them to define personalized labels, including setting the number of labels (between 2 and 5) and customizing their names and offers checkboxes to

control whether information extraction and automatic response generation should be performed. Additional style options, such as polite or formal tone, are also available to tailor the assistant's output. The batch operation mode extends this functionality for efficiency. Instead of manual input, users specify the number of recent emails to process, and the system automatically retrieves them via Google Mail integration. It retains the same customization features, such as personalized labels, extraction, and response generation, enabling consistent policy application across multiple emails at once. Both modes include visual feedback areas for results and system status, and support operations such as copying outputs or clearing processed data.

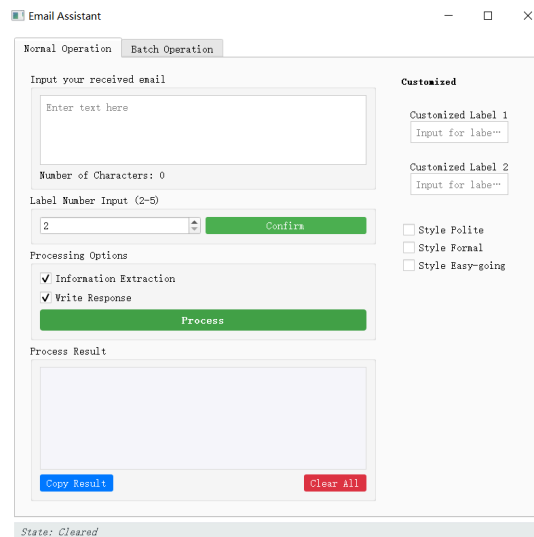


Fig. 3. Normal Operation layout

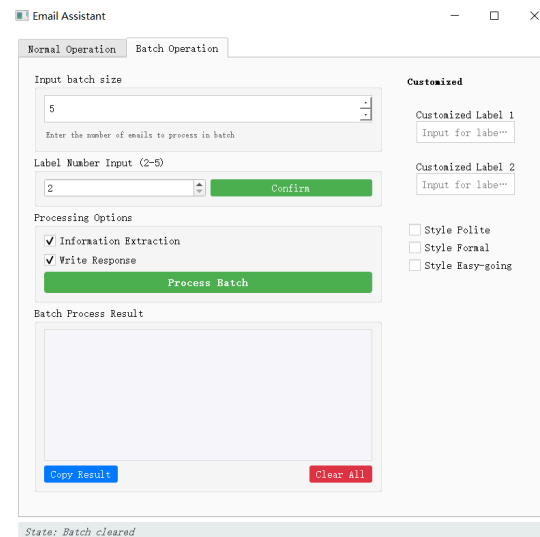


Fig. 4. Batch Operation layout

For testing, Fig. 5 is an example of normal operation. We can tell from the result that the classification task was completed based on user-defined labels. Similar to that, Fig. 6 shows the result of batch operation.

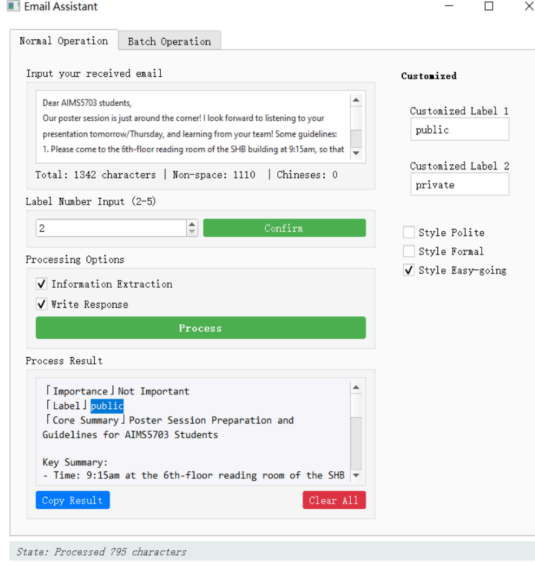


Fig. 5. Normal Operation example

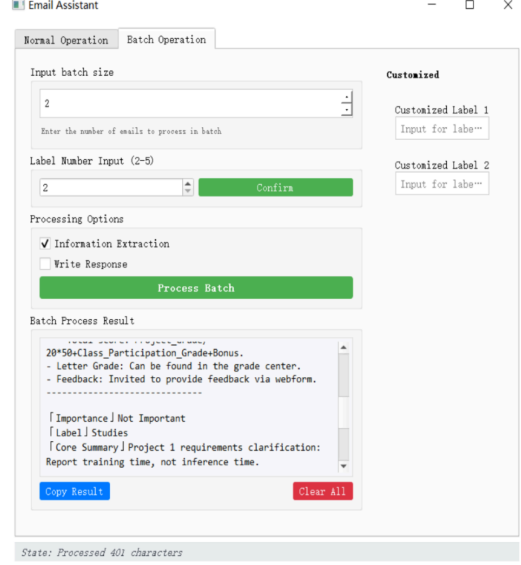


Fig. 6. Batch Operation example

5. Discussion

The development of our personalized email assistant demonstrates that a precisely architected system, leveraging reasoning enhancement techniques, can effectively overcome the limitations of lightweight LLMs. This discussion interprets our key findings, examines the broader implications of our approach, handles the system's limitations, and indicates pathways for future work.

Our core premise was that a personalized email assistant in school content could be built without relying on large proprietary models or extensive fine-tuning parameters. The results validate that this is achievable through a synergistic combination of a multi-agent pipeline, Chain-of-Thought prompting, and Retrieval-Augmented Generation.

The parallel multi-agent architecture was useful in decomposing the complex task of email classification into tractable subtasks. This method for well-defined application pipelines, task decomposition is a more effective strategy for lightweight models than relying on a single prompt.

The distinct and complementary roles of CoT and RAG methods are clearly evidenced in our experiments. CoT significantly improved classification accuracy and intent discernment by analysing the output of Table 1. It also aligns the model's output more closely with human contextual understanding. And RAG primarily enhanced the precision and specificity of information extraction. By grounding the model's generation in retrieved campus-specific knowledge, RAG transformed LLM's role from a generic summarizer to a precise information extractor. For lightweight models, external knowledge retrieval serves not only

as a supplementary method but also as an essential prerequisite for performing detailed, domain-specific extraction tasks. This approach delivers on our promises of personalization, privacy, and low cost goals. By operating entirely on-device or within a self-hosted environment using a compact model like ERNIE-Speed, we eliminate API costs and data privacy risks associated with cloud services, addressing a significant gap in current commercial offerings.

The success of CoT in reclassifying emails such as from "advertisement" to "study" indicates that the primary bottleneck for classification was not a lack of knowledge, but a deficiency in systematic reasoning. Providing a step-by-step reasoning scaffold effectively unlocked the model's latent capacity to make nuanced judgments, reducing overfitting to salient but misleading terms like "opportunity" or "announcement." The integration of precise, retrieved context enabled it to perform like a domain-aware agent. This suggests that for small-scale and low-cost applications, the value of a tightly selected, compact knowledge base can outweigh the benefits of a larger, more general-purpose model.

Despite promising results, our work has several limitations that must be acknowledged. First, there is an inherent ceiling to the reasoning capability of a 10-15B parameter model. While CoT provides a scaffold, it cannot endow the model with deep, abstract reasoning or genuine comprehension of highly complex, ambiguous, or intentionally deceptive emails. Performance on edge cases will likely remain inferior to that of large-scale models. Second, the system's effectiveness is contingent on the quality and maintenance of the external knowledge base. While we reduced data dependency for training, we introduced a dependency on curated domain knowledge. The process of constructing and updating this knowledge base, though smaller in scale than collecting thousands of labeled emails, still requires manual effort and lacks full automation. Third, our evaluation, while illustrative, is not exhaustive. A more rigorous assessment would require a large-scale, labeled dataset of school emails to report standard metrics like precision and recall across all categories. The current qualitative and example-based analysis, though compelling, limits direct comparability with other systems. Finally, the rule execution component, while demonstrated for basic actions, may face challenges with complex, multi-step workflows involving external APIs or requiring nuanced state management. Ensuring the robustness and security of automated actions remains an area for further engineering.

Based on this foundation, several future directions appear. A logical next step is to explore hybrid adaptation strategies. Techniques like Low-Rank Adaptation could be used for efficient fine-tuning on a small corpus of user-corrected labels or extraction formats, potentially improving performance further without sacrificing efficiency or privacy, moving beyond a purely prompt-based personalization. Future work will involve adapting this multi-agent, CoT+RAG framework to other domains requiring personalized text processing, such as corporate ticket routing or community moderation, to validate its generalizability. Incorporating user feedback is crucial for sustained improvement. Allowing users to easily correct misclassifications or extraction errors, and using this signal to subtly adjust prompts

or re-rank retrieval results, would transform the system from a static tool into a continuously learning personal assistant.

6. Conclusion

This research aimed to solve the problem of email overload in academic environments by creating a personalized and cost-effective intelligent assistant. By strategically combining a multi-agent architecture with CoT prompting and RAG, we enabled a lightweight LLM that runs locally to handle complex email processing tasks. These tasks include multi-label classification and structured information extraction, all achieved well.

More broadly, our work presents a practical case for an alternative approach to AI development. Instead of depending only on ever-larger models, this paradigm focuses on smart engineering and system design built around efficient models. This strategy makes advanced AI assistance more accessible, easier to deploy, and more trustworthy. It thereby enables wider adoption in settings where privacy and resources are key concerns, such as in education, business, and personal computing.

References

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AI Usage Declaration: In this project, we used Deepseek to assist with refining the text in the Introduction and conclusion section, and for debugging the GUI code. All core logic and experimental design were conducted by the human authors.